

COEN/ELEC 390
Computer and Electrical Engineering Product Design Project
Electrical and Computer Engineering
Summer 2026

Course Instructor:

Dr. Wahab Hamou-Lhadj
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Office Hours:

Tuesdays from 2:30pm to 3:30pm EV 5.213

Lectures:

COEN / ELEC 390-BB - Tu Th 1:15PM - 2:30PM Room: MB S1.235 SGW

Tutorials:

COEN 390-BB A / ELEC 390-BB A Tu 2:45PM - 4:25PM
COEN 390-BB B / ELEC 390-BB B We 2:45PM - 4:25PM

COEN 390/ELEC 390 Project Specialist:

- **Bipin Patel (pbipin@encs.concordia.ca). Room H851-1.**

Labs:

Laboratory Equivalent time, 6 hours per week.

Course Calendar Description:

The Product Design Project reinforces skills introduced in ENGR 290, which include teamwork, project management, engineering design for a complex problem, technical writing, and technical presentation in a team environment. It also introduces students to product development. Students are assigned to teams and each team develops, defines, designs and builds a system and/or device under broad constraints set by the Department. Students present their product definition and design, and demonstrate that their system/device works at the end of the term.

Prerequisites:

COEN 390 Prerequisites : COEN 311, COEN 352; ENGR 290. Students must complete a minimum of 45 credits in the BEng (Computer) prior to enrolling.

ELEC 390 Prerequisites: COEN 352; ELEC 311; ENGR 290, Students must complete a minimum of 45 credits in the BEng (Electrical) prior to enrolling.

Co-requisites: COEN/ELEC 490 Capstone Design Project

Specific Knowledge and Skills Needed for this Course:

Students taking this course are expected to have sufficient knowledge of the following topics. Should you have difficulties in any of these topics, you are strongly encouraged to review them before the DNE deadline.

- Programming (COEN 243, COEN 244, and COEN 352)
- Computer architecture (COEN 311, ELEC 311)
- Design process (ENGR 290)

Course materials:

Required Textbook: Essential Scrum, Kenneth S. Rubin, Addison-Wesley, 2013.

Not required, but useful:

- Product Design and Development, K.T. Ulrich, S.D. Eppinger, McGraw-Hill Irwin, 7th edition, 2019.
- Managing Software Requirements, a Use Case Approach by Dean Leffingwell & Don Widrig, Addison-Wesley, 2003.

Grading Scheme

Assignments	15%
Milestone 1 – Mission statement	10%
Milestone 2 – Requirements document	15%
Sprint 1	
Pre-sprint planning	1.50%
Demo of potentially shippable code	5%
Documentation	3.50%
Sprint 2	
Pre-sprint planning	1.50%
Demo of potentially shippable code	5%
Documentation	3.50%
Sprint 3	
Pre-sprint Planning	1.50%
Final document submission	10%
Final demo	10%
Final presentation	3.50%
Excellence	10%
Individual	5%

Notes:

- The excellence grade will be based on the user experience, the creativity in conception and execution of the mission, and the excellence of the elicitation and adherence to user needs.
- Some grades may be awarded or taken away based on individual effort.

Tentative Course Schedule

The lectures begin in the first week of the term. The main lectures will be over after 5 weeks, but we will still meet afterwards on a regular basis to discuss the project.

Week	Lecture
Week 1	1. Course description 2. Development processes & organizations 3. Product Planning
Week 2	4. Understanding users and stakeholder needs / conducting interviews 5. Essential Scrum
Week 3	6. Agile Principles 7. Sprints Planning, Execution, and Retrospective 8. Requirements Gathering
Week 4	8. Product Backlog 9. Estimation and Velocity
Week 5	10. Product Owner 11. Scrum Master 12. Development Team

Engineering Tools

- Android Development Studio (<https://developer.android.com/studio>)
- Code versioning system (e.g., Github)
- Microcontrollers and various hardware sensors

Details on assessment tools:

A. Assignments:

There will be two assignments to practice Android development.

B. Project description:

The objective of the project is to design and build a system and/or device of your choice, which combines an Android app, sensors/actuators, and basic data analytic capabilities. An example of a project would be the development of a “stress prompter app”, which measures heart rate and potentially other body signals and prompt the user to do stress reducing exercises when appropriate. In this project, you will need to follow sound principles of project management, product development, and engineering design for complex problems. The project also puts strong emphasis on team work, technical writing and presentation, and ethics and equity. More details about the project will be posted in a separate document on Moodle.

Other information

N/A

Graduate Attributes:

Graduate Attributes	Indicator	Assessment Tool	CLOs
Design. An ability to design solutions for complex, open-ended engineering problems and to design systems, components or processes that meet specified needs with appropriate attention to health and safety risks, applicable standards, and economic, environmental, cultural and societal considerations.	Define the objective	Milestone 2 and sprints	a, b
	Idea generation and selection	Milestone 1 and sprints	a, c
	Detailed design	Sprints and Final submission	c
	Validation and implementation	Sprints and final submission	d, e
Ethics and equity: An ability to apply professional ethics, accountability and equity	Professional ethics and accountability	Milestone 2	f
	Equity	Final submission	f

Course Learning Outcomes (CLOs):

By the end of this semester, students are expected to master the following computer/electrical engineering concepts.

- a) Elicit and document the requirements of a customer
- b) Generate a product specification in the product backlog/story format.
- c) Design and code a sensor enabled Android App, using the scrum design methodology.
- d) Carry out basic software testing and software documentation
- e) Participate and possibly lead a small engineering team
- f) Understand the ethical dimensions and issues surrounding the designed App
- g) Prepare and present written documentation
- h) Prepare and present oral presentations to colleagues

Health and Safety Guidelines

All health and safety rules specific to this course can be found in the lab manual. General health and safety instructions and available health and safety trainings can be found at:

[Safety Programs - Concordia University \(https://www.concordia.ca/campus-life/safety/general-safety.html\)](https://www.concordia.ca/campus-life/safety/general-safety.html)

[If your course has additional information about health and safety guidelines/training, please insert them here.](#)

On Campus Resources

Please visit [Student services at Concordia University](#) for the services available Gina Cody School students.